

MICRO-ECONOMICS:

--> Economics is the social science that studies how individuals, businesses, governments, and societies make choices about allocating limited resources to satisfy their needs and wants.

--> **CPI** stands for **Consumer Price Index**, which is a measure that examines the average change over time in the prices paid by consumers for a basket of goods and services. It is often used as an indicator of inflation.

--> **Difference between Micro and Macro Economics:**

Microeconomics: Focuses on individual and small-scale economic activities. It examines the decision-making processes of individuals, households, and firms, and how they interact in specific markets. It looks at the behavior of buyers and sellers, pricing, production, and consumption of goods and services on a small scale.

Macroeconomics: Deals with the economy as a whole. It looks at large-scale economic factors like national income, total employment, inflation, and government policies. It studies how different sectors of the economy are interconnected and how government policies can influence overall economic growth, stability, and inflation.

--> **Principles of Economics:**

1. **Trade off:** A trade-off refers to a situation where choosing one option means giving up another. It reflects the concept that resources (time, money, energy, etc.) are limited, and using them for one purpose requires forgoing their use for something else.

2. **Cost of choices:** The cost of choices refers to the concept that every decision involves a cost, often in terms of time, money, effort, or missed opportunities. This is closely related to another term called Opportunity cost. ("This is the most significant cost in decision-making. It's what you give up when you choose one option over another. For example, if you decide to spend money on a new phone, the opportunity cost could be the other things you could have bought with that money (like savings or travel).")

3. **Marginal principle:** It states that rational decision-makers will take action only if the marginal benefit of the action exceeds the marginal cost.

4. People responds to incentives during voting, etc.

Structures of Market:

Monopoly and Oligopoly are two distinct market structures that describe how firms interact within a market. Each has its own characteristics and implications for competition, pricing, and consumer choice.

Monopoly:

A monopoly is a market structure where **a single firm or entity is the sole provider of a particular good or service**. This firm has significant control over the market and can influence prices and output levels.

Key Features:

1. **Single Seller**: There is only one firm that supplies the product or service, meaning **no direct competition**.
2. **Price Maker**: The monopolist can **set prices because they are the only supplier**. This often leads to **higher prices and reduced output compared to competitive markets**.
3. **Barriers to Entry**: High barriers prevent other firms from entering the market. **Barriers can include high startup costs, exclusive access to essential resources, or government regulations**.
4. **Lack of Substitutes**: The product or service provided by the monopolist often has **no close substitutes, making consumers reliant on the single provider**.

Examples:

- **Utilities**: Local utilities such as electricity and water services often operate as monopolies due to **high infrastructure costs and regulatory constraints**.
- **Patented Pharmaceuticals**: Companies with exclusive patents on drugs may have a **monopoly on that medication** until the patent expires.

Oligopoly:

An oligopoly is a market structure where **a small number of firms dominate the market**. These firms hold a **large share of the market and their decisions impact each other, leading to strategic interactions**.

Key Features:

1. **Few Sellers:** The market is controlled by a limited number of firms, which means each firm's actions affect the others.
2. **Interdependence:** Firms in an oligopoly must consider the potential reactions of their rivals when making decisions about prices, output, and other strategic variables.
3. **Barriers to Entry:** There are significant barriers to entry, such as **high capital requirements, economies of scale, or strong brand loyalty**, which prevent new competitors from entering the market easily.
4. **Product Differentiation:** Products may be either similar (e.g., oil companies) or differentiated (e.g., car manufacturers), depending on the industry.

Examples:

- **Automobile Industry:** Major car manufacturers like **Toyota, Ford, and General Motors** dominate the market, influencing each other's pricing and product offerings.
- **Telecommunications:** **Few large firms often control the market for phone and internet services**, leading to competitive but interdependent behavior.

Quantity Demand (Q_d): It is the function of **price (p), income (m), price of related goods (P_r) and taste & preferences of consumers ($T \& P$)**. Change in anything changes demand curve.

Law of demand: "All else being equal, as the price of a good or service decreases, the quantity demanded for it increases; conversely, as the price increases, the quantity demanded decreases."

Demand based on income (m): Basically, it is categorized in two types: Normal goods and Inferior goods.

Normal goods are goods for which demand **increases as consumer income rises**, and decreases when income falls. Examples: clothing, electronics, dining out, etc. These have positive relationship. **Demand Curve:** For normal goods, **an increase in income shifts the demand curve to the right** (indicating higher demand at each price level), while a decrease in income shifts it to the left.

Inferior goods are goods for which **demand decreases as consumer income rises**, and **increases when income falls**. Examples: Low-cost noodles, public-transport, etc. They have negative relationship. **Demand Curve:** For inferior goods, **an increase in income shifts the demand curve to the left** (indicating lower demand at each price level), while **a decrease in income shifts it to the right**.

Quantity Supply (Qs): It is the function of price(P), raw material price($p_{i/p}$), rent (R), workers(W), and technology(T).

ELASTICITY: Elasticity refers to the measure of how responsive one variable is to a change in another variable. Most commonly, it describes how much the quantity demanded or supplied of a good change in response to a change in its price, income, or the price of related goods.

Price Elasticity of Demand (PED):

- **Definition:** Measures how much the quantity demanded of a good change when its price changes.
- **Example:** If a small change in the price of a luxury car leads to a large drop in sales, the demand for that car is said to be **price elastic**.

Three main categories of elasticity:

1. Elastic:

- **Definition:** A good is said to be **elastic** if a small change in price leads to a relatively larger change in the quantity demanded or supplied.
- **Characteristics:**
 - **Elasticity value:** Elasticity is greater than 1 ($E > 1$).
 - Consumers are highly sensitive to price changes.
- **Example:** Luxury goods, such as designer clothing or high-end electronics. If the price of a luxury watch increases by 10%, the quantity demanded might drop by 20%, indicating that demand is elastic.
- **Effect:** When the price increases, the total revenue decreases, as the drop in quantity demanded is larger than the price increase.

2. Inelastic:

- **Definition:** A good is said to be **inelastic** if a change in price leads to a relatively smaller change in the quantity demanded or supplied.
- **Characteristics:**
 - **Elasticity value:** Elasticity is less than 1 ($E < 1$).
 - Consumers are less sensitive to price changes.
- **Example:** Necessities such as insulin for diabetics or gasoline. If the price of gasoline increases by 10%, the quantity demanded might decrease by only 2%, meaning demand is inelastic.

- **Effect:** When the price increases, total revenue increases, as the change in quantity demanded is relatively small compared to the price increase.

3. Unit Elastic:

- **Definition:** A good is said to be **unit elastic** when the percentage change in price leads to an equal percentage change in quantity demanded or supplied.
- **Characteristics:**
 - **Elasticity value:** Elasticity is equal to 1 ($E = 1$).
 - The percentage change in quantity demanded is exactly equal to the percentage change in price.
- **Example:** A product where a 10% increase in price leads to a 10% decrease in quantity demanded.
- **Effect:** When price changes, total revenue remains unchanged because the proportionate change in quantity offsets the price change.

Perfectly elastic demand or supply refers to an extreme case of elasticity, where even a very small change in price causes an infinite change in the quantity demanded or supplied.

Perfectly inelastic demand or supply refers to a situation where the quantity demanded or supplied remains unchanged regardless of any change in price.

Income elasticity of demand (YED) measures how the quantity demanded of a good changes in response to changes in consumers' income.

Formula:

Income elasticity demand (YED) = $\frac{\% \text{ change in quantity demanded}}{\% \text{ change in income}}$

- If $YED > 1$, demand is **income elastic**, meaning demand changes more than the change in income.
- If $YED < 1$, demand is **income inelastic**, meaning demand changes less than the change in income.

Gross price elasticity of demand (XED) refers to how the quantity demanded of one good changes in response to a change in the price of another related good.

Formula:

$XED = \frac{\% \text{ change in quantity demanded of good 1}}{\% \text{ change in price of good 2}}$

- **XED > 0:** The goods are substitutes.
- **XED < 0:** The goods are complements.

Elasticity of supply measures the responsiveness of the quantity supplied of a good or service to a change in its price. It shows how easily producers can increase or decrease production in response to changes in price.

Consumer surplus refers to the difference between what consumers are willing to pay for a good or service and what they actually pay.

Formula for Consumer Surplus (Using Integration):

$$\text{Consumer Surplus} = \int_0^Q D(q) dq - P \times Q$$

- $D(q)$ is the demand function, representing the price consumers are willing to pay for a given quantity q .
- P is the market price.
- Q is the quantity demanded at the market price.

Producer surplus is the difference between the price at which producers are willing to sell a good or service and the price they actually receive.

Formula for Producer Surplus:

$$\text{Producer Surplus} = P \times Q - \int_0^Q S(q) dq$$

Where:

- P is the market price.
- Q is the quantity supplied at the market price.
- $S(q)$ is the supply function, representing the minimum price at which producers are willing to sell a given quantity q .